UNIVERSITY OF BOHOL

COLLEGE OF ENGINEERING, TECHNOLOGY,

ARCHITECTURE, AND FINE ARTS

A Report for

CPEP112

Final Project

Presented to the Dean and Faculty of College of Engineering,

Technology, Architecture, and Fine Arts

University of Behol

In Partial Fulfillment of the Requirements fer the Degree of

Bachelor of Science in Computer Engineering

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"Pharmacy Lab Inventory System"

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BSCompE – 1

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Page	Section Title
1	System Overview – Pharmacy Laboratory Inventory System
1	System Functionalities
1	Admin Functions
1	Student Functions
2	Database Schema Overview
2	Database Tables Description
3	Sample Run of the System
4	Class Diagram (Text Format)
4	Java Classes and Their Descriptions
5	Group Details – Group Name and Meaning
5	Members and Responsibilities
6	Members' Curriculum Vitae – Marc Kenneth A. Alita
7	Members' Curriculum Vitae – Cris Rodolfo D. Yu
8	Members' Curriculum Vitae – Justine Besagas Juyad
9	Client Information
10	Student Sign Up (PHP)
11	Student Log In (PHP)
12	Student Dashboard (PHP)
13	Student Lab Inventory View (PHP)
14	Student Borrowing Item (PHP)
15	Student Pending Request (PHP)

16	Student Borrowing History (PHP)
17	Lab Assistant Sign Up (PHP)
18	Lab Assistant Log In (PHP)
19	Lab Assistant Dashboard (PHP)
20	Manage Inventory (PHP)
21	Manage User (PHP)
22	Add User/ Show Existing User (PHP)
23	Manage Borrowing (PHP)
24	Return Process (PHP)
25	LOG IN / Invenotry (GUI Java)
26	Borrowing Hisotry (GUI Java)
27	Checking Records Screenshot
29	Group Journal During Project Development
30	Presenting the System to Client(YouTube Video URL)

System Description

The Pharmacy Laboratory Inventory System is a system designed to manage laboratory tools and equipment used in pharmacy classes. It allows students to request items and view their borrowing history, while administrators can manage inventory, approve requests, and generate logs. The system uses PHP and MySQL for the web interface and Java for GUI-based admin tasks.

System Functionalities

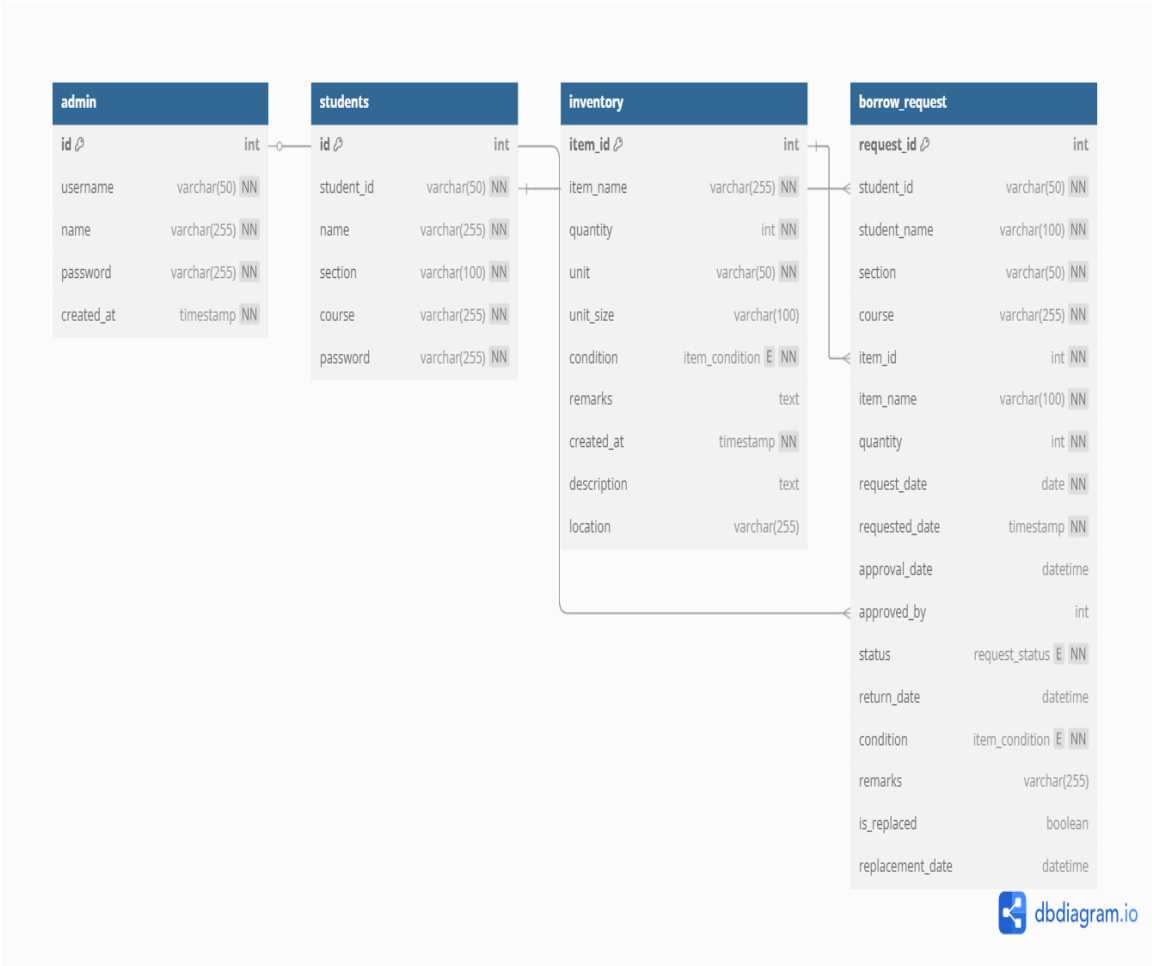
Admin Functions

- Login (Java GUI or Web)
- Add/Edit/Delete apparatus
- View, approve, or reject borrow requests
- Update inventory counts
- Monitor borrowing history

Student Functions

- Sign up and log in (Web)
- Borrow apparatus via form
- View request status
- Access borrowing records

Database Schema Overview



- **admins**

- Stores administrator login details
- Likely contains fields like admin_id, username, password, fullname

- **borrow_request**

- Stores borrowing transactions by students
- Possible fields: request_id, student_id, item_id, date_requested, status

- **inventory**

- Core table for apparatus/inventory details
- Possible fields: inventory_id, item_name, quantity, location, status

  - **item**

- Might be used for referencing item details (possibly duplicates some fields in inventory)
- Fields might include item_id, description, category

  - **manage_user**

- Manages normal users/students who can borrow items
- Likely fields: user_id, fullname, username, password, role

  - **students**

- Detailed student info (maybe linked to borrow_request)
- Likely fields: student_id, name, section, course, year_level

Sample Run of the System

1. Student logs in via PHP interface.
2. Submits a borrow request for apparatus.
3. Admin logs in via Java GUI.
4. Admin approves or rejects the request.
5. Inventory updates automatically.
6. Student sees updated status in their account.
7. All transactions are saved in the database.

Class Diagram (Text Format)



This diagram shows that all logic classes (LoginSystem, BorrowSystem, InventorySystem) communicate with the central DatabaseSystem, which handles the connection to MySQL tables like inventory and borrow_request.

Java Classes and Their Descriptions (Quick Stat Tool)

Class Name	Purpose
LoginSystem	Handles admin login logic. Connects to the database and validates credentials using the admins table.
InventoryChecker	Fetches all apparatus from the inventory. Identifies and flags items with low stock (e.g., quantity ≤ 5).
BorrowingHistory	Displays all approved and returned borrowing requests. Shows request details such as item name, quantity, student ID, date, and status.

Group Details

Group Name: MADUDA

Meaning: MADUDA is our unique group name formed by combining the first two letters from each of our hometowns - "MA" from Maribojoc, "DU" from Duero, and "DA" from Dauis. We chose this name because it perfectly represents our unity while honoring our individual roots, blending Maribojoc's historic strength, Duero's flowing resilience, and Dauis' coastal serenity into one powerful identity that symbolizes how different backgrounds can come together to create something special and uniquely ours.

Members and Responsibilities

Name	Role	
Marc Kenneth Alita	Frontend, Backend Developer	Documents, HTML
Justine Juyad	Frontend, Backend Developer	HTML, CSS, UI/UX
Cris Rodulfu Yu	Frontend, Backend Developer	JAVA, Report Writing

Members Curriculum Vitae

Personal Profile



Name: Marc Kenneth Asedilla Alita

Address: Dob-dob, Dipatlong, Maribojoc, Bohol

Birth Date: October 23, 2005

Age: 19

Birth Place: Community Hospital Tagbilaran City

Citizenship: Filipino

Civil Status: Single

Father's Name: Gil D. Alita

Mother's Name: Maricel A. Alita

Email: mkaalita@universityofbohol.edu.ph

EDUCATIONAL BACKGROUND

Elementary: Santa Cruz Elementary School

Secondary: Holy Spirit School of Tagbilaran

Tertiary: University of Bohol Course : Bachelor of Science of Computer Engineering

Personal Profile



Name: Cris Rodolfo D. Yu

Address: Brgy. Mayacabac, Dauis, Bohol

Birth Date: February 21, 2004

Age: 21

Birth Place: Cebu City

Citizenship: Filipino

Civil Status: Single

Father's Name: Jose Zaldy J. Yu

Mother's Name: Salina C. Yu

Email: crisrodolfod.yu95@gmail.com

EDUCATIONAL BACKGROUND

Elementary: Claver National High

School Secondary: Claver National High School

Tertiary: University of Bohol Course : Bachelor of Science of Computer Engineering

Personal Profile



Name: Justine Besagas Juyad

Address: Bangwalog Duero Bohol

Birth Date: January 25 , 2006

Age: 19

Birth Place: Bangwalog Duero Bohol

Citizenship: Filipino

Civil Status: Single

Father's Name: Eddie Juyad

Mother's Name: Maria Juyad

Email: itsmejustinejuyad@gmail.com

EDUCATIONAL BACKGROUND

Elementary: Bangwalog Duero Bohol

Secondary: Central Visayan Institute Foundation (CVIF)

Tertiary: University of Bohol Course: Bachelor of Science of Computer Engineering

Client Information

1. **Dr. Sofia Lopez** *(Dean of Pharmacy)*

- **Role:** Dean / Approving Authority
- **Institution:** University of Bohol
- **Project Role:**
    - Approved the system (but admitted she "has no idea how it works").
    - Supported the idea to reduce paperwork.
- **Client Description:**
    - Signs off on lab processes but relies on her assistant for details.
    - Wants a system so she doesn’t have to deal with manual reports.

2. Mr. Carlo Mendoza *(Laboratory Assistant)*

- **Role:** Assistant / Lab Coordinator
- **Institution:** University of Bohol
- **Project Role:**
    - Told us they do everything **manually** (tracking with paper logs).
    - Asked for a system to **stop the burden** of lost/delayed records.
- **Client Description:**
    - Handles the actual equipment tracking.
    - Struggles with missing items and messy paperwork.

Example Run in PHP

STUDENT SIDE

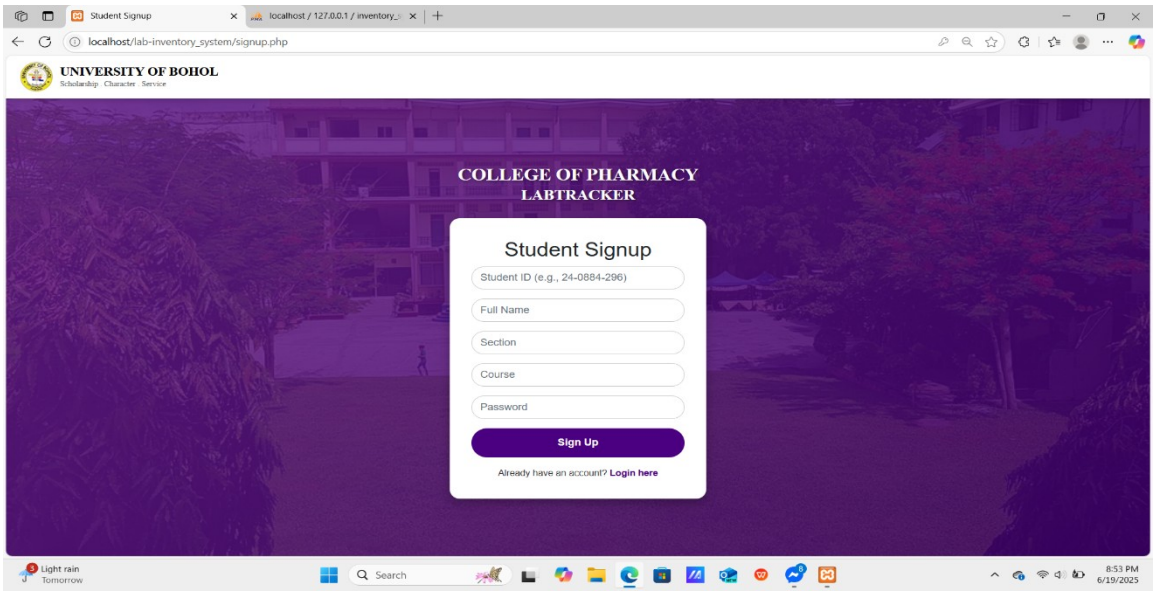
1. Student Sign Up

Description:

Allows new students to register an account by providing their full name, ID number, section, course, email, and password.

Fields:

- **Full Name**
- **Student ID**
- **Section**
- **Course**
- **Password**



2. Student Login

Description:

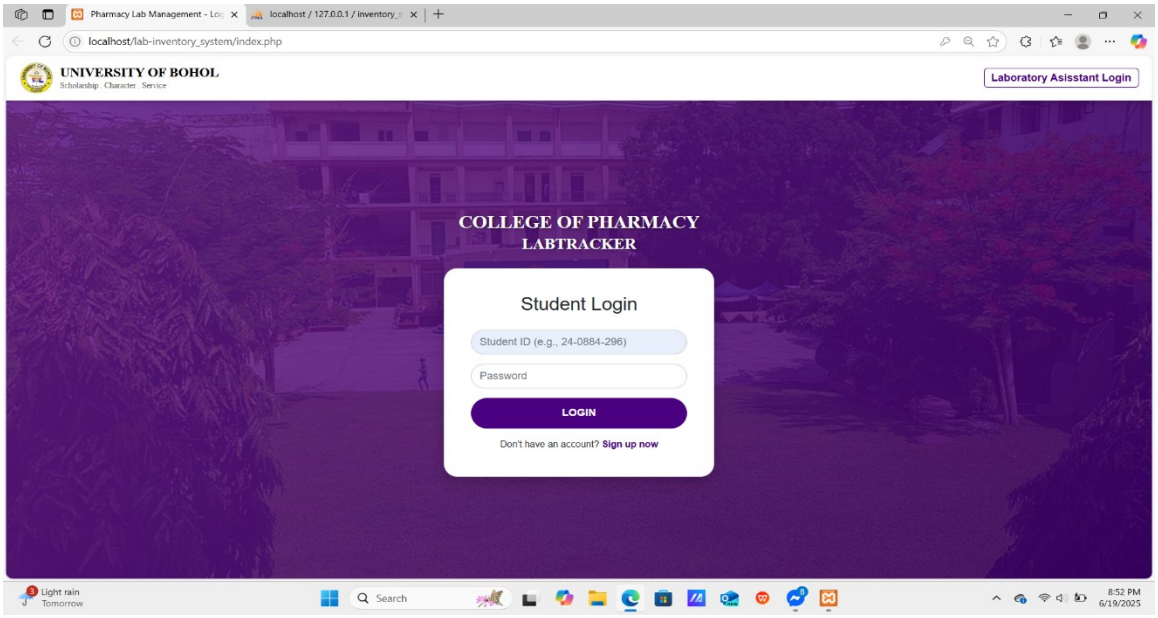
Registered students can log in using their ID and password to access the dashboard.

Fields:

- **ID no.**
- **Password**

Validation:

Checks credentials and redirects to student dashboard upon success.

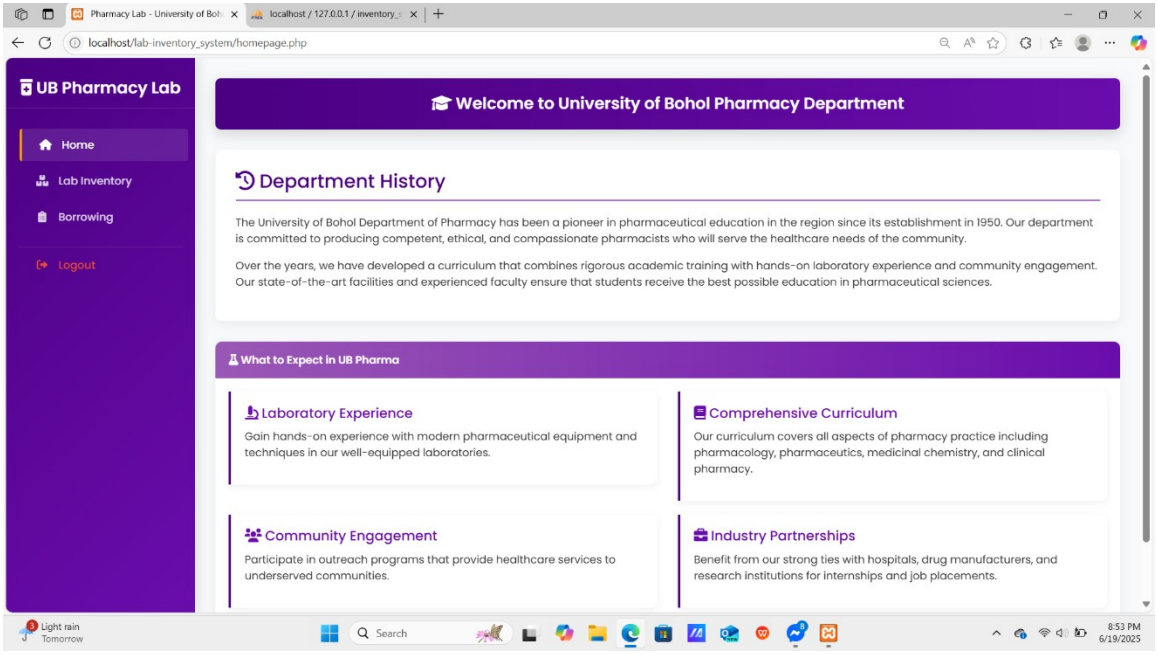


3. Student Dashboard

Description:

Main page showing student navigation options:

- **View available inventory**
- **Submit a borrow request**
- **Track borrowing status and history**
- **Logout**



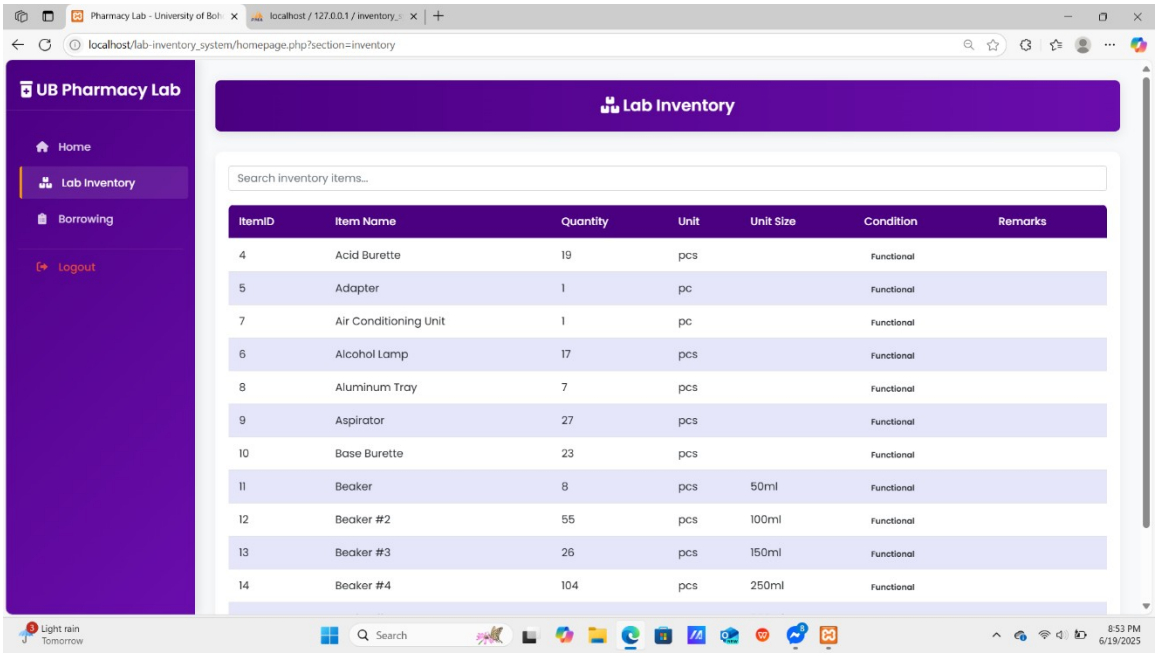
4. Lab Inventory View

Description:

Displays a table of all inventory items that students can borrow.

Includes:

- **Item Name**
- **Available Quantity**
- **Unit (e.g., pcs, sets)**
- **Condition (New, Good, Damaged)**
- **Storage Location**



5. Borrow Item

Description:

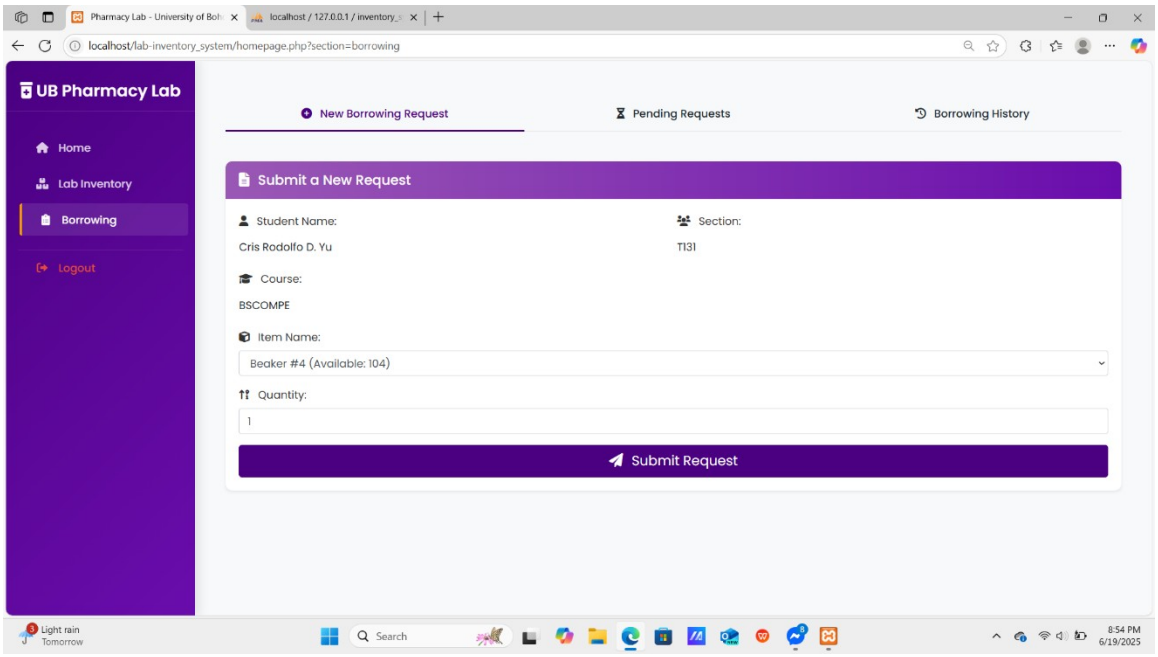
Students fill out a form to request an item from the lab.

Fields:

- **Item to borrow (dropdown list)**
- **Quantity**
- **Student details auto-filled from login**

Functionality:

- **Adds a new entry to the borrowing requests**
- **Sets status as “Pending”**



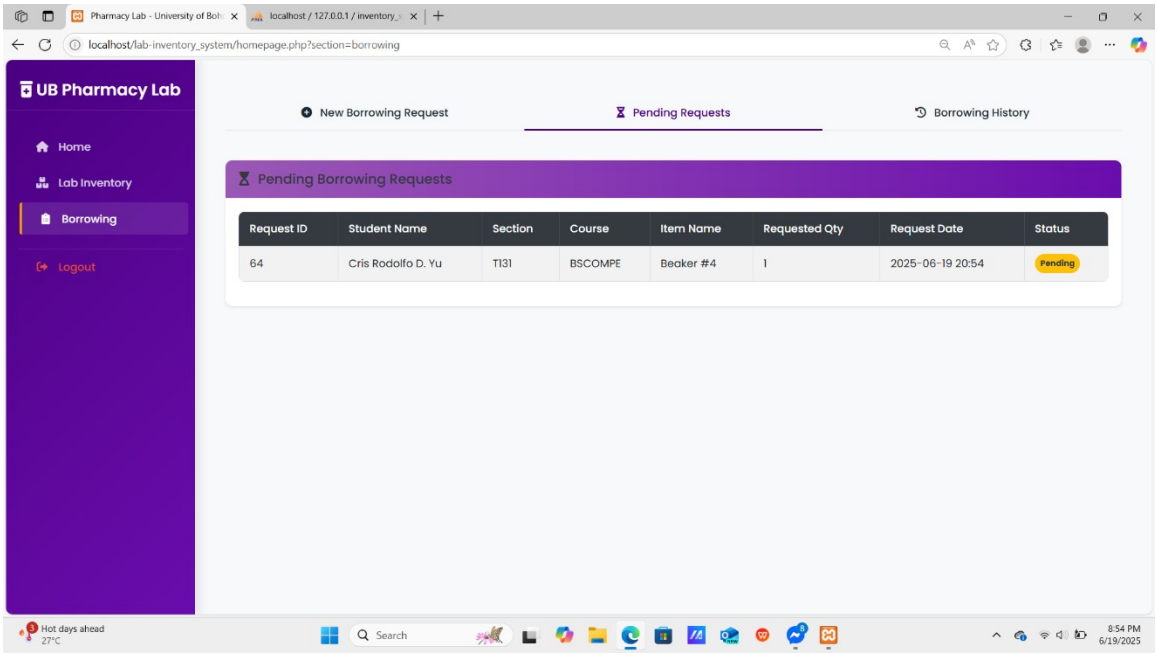
6. Pending Requests

Description:

Shows all borrowing requests that have not yet been approved or declined by the admin.

Displayed Fields:

- **Item Name**
- **Requested Quantity**
- **Request Date**
- **Current Status (Pending)**



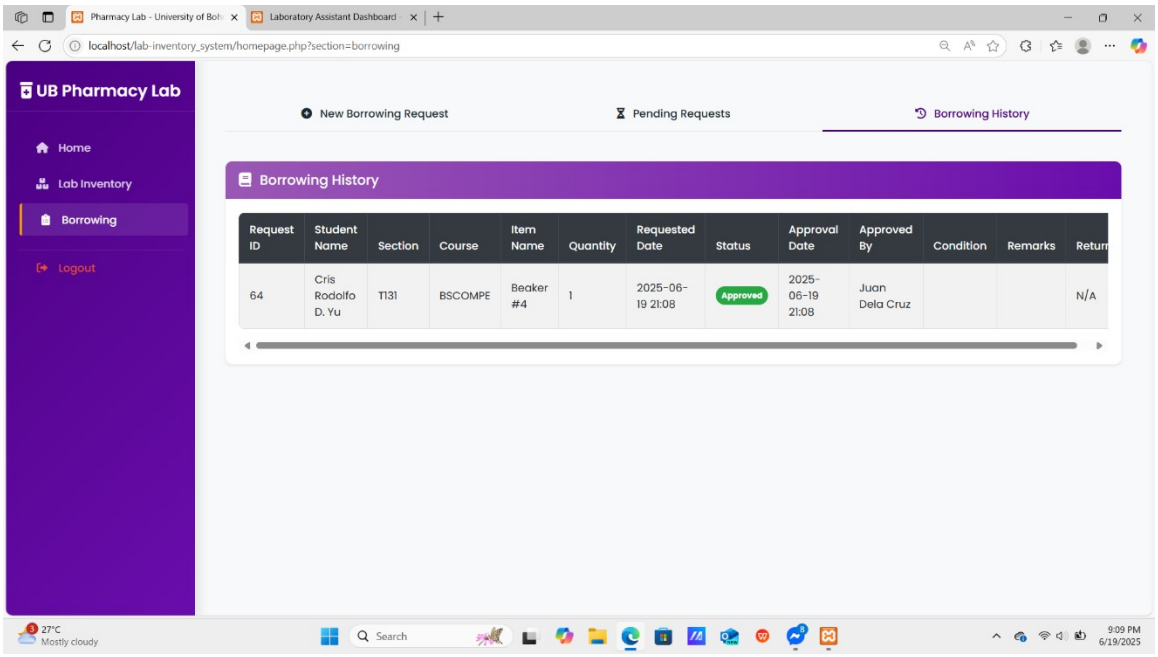
7. Borrowing History

Description:

Lists all past and completed borrowing activities by the student.

Includes:

- **Item Name**
- **Date Borrowed**
- **Date Returned**
- **Status (Approved, Returned, Declined)**



ADMIN SIDE (Laboratory Assistant)

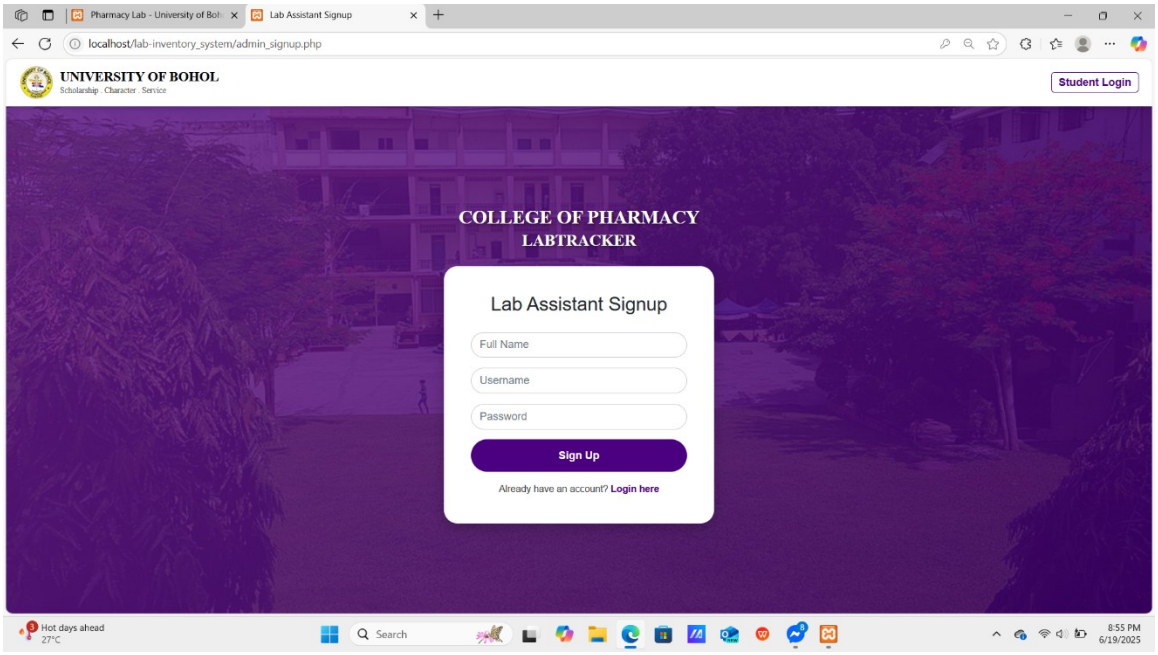
1. LAB ASSISTANT Sign Up

Description:

Allows new lab assistant to register an account by providing their full name, username and password.

Fields:

- **Full Name**
- **username**
- **Password**



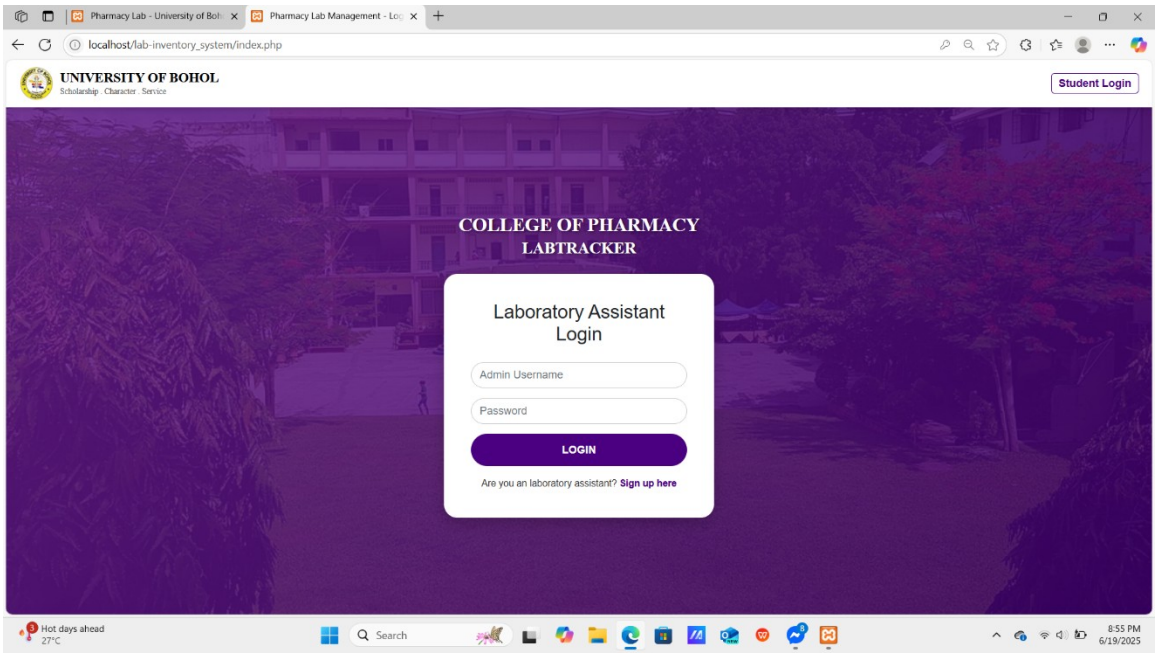
2. LAB ASSISTANT Login

Description:

Admins or laboratory assistants log in using their credentials to manage the system.

Fields:

- **username**
- **Password**

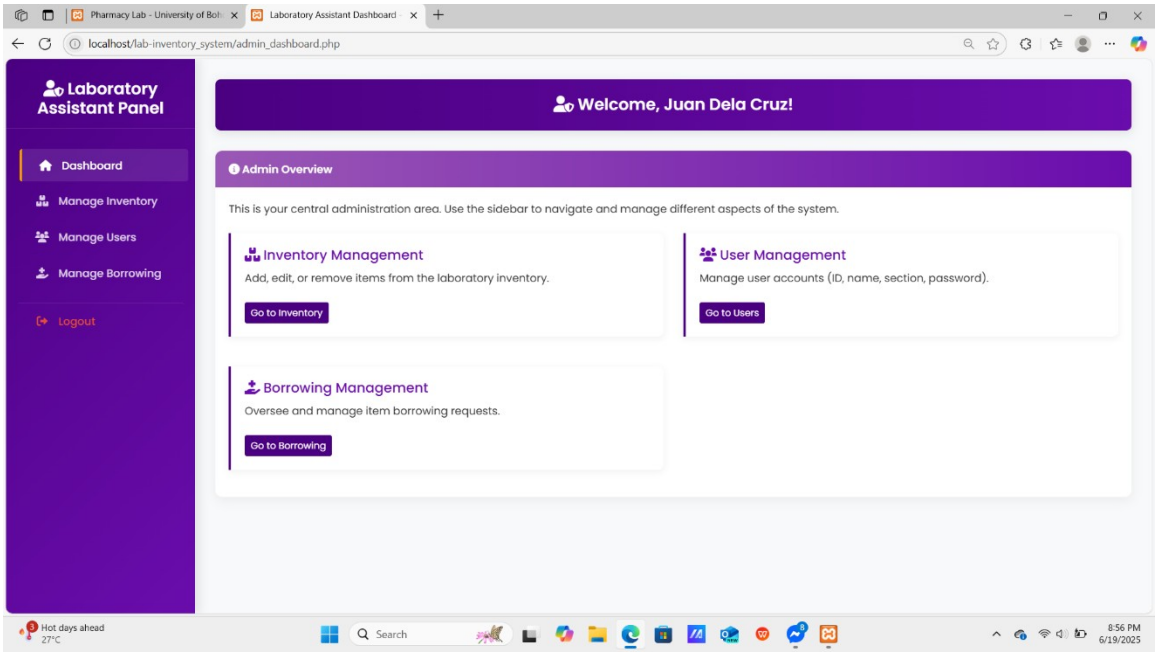


3. LAB ASSISTANT Dashboard

Description:

Central page for admin functions, with quick access to:

- **Inventory management**
- **User management**
- **Borrow request management**
- **Logout**



4. Manage Inventory

Description:

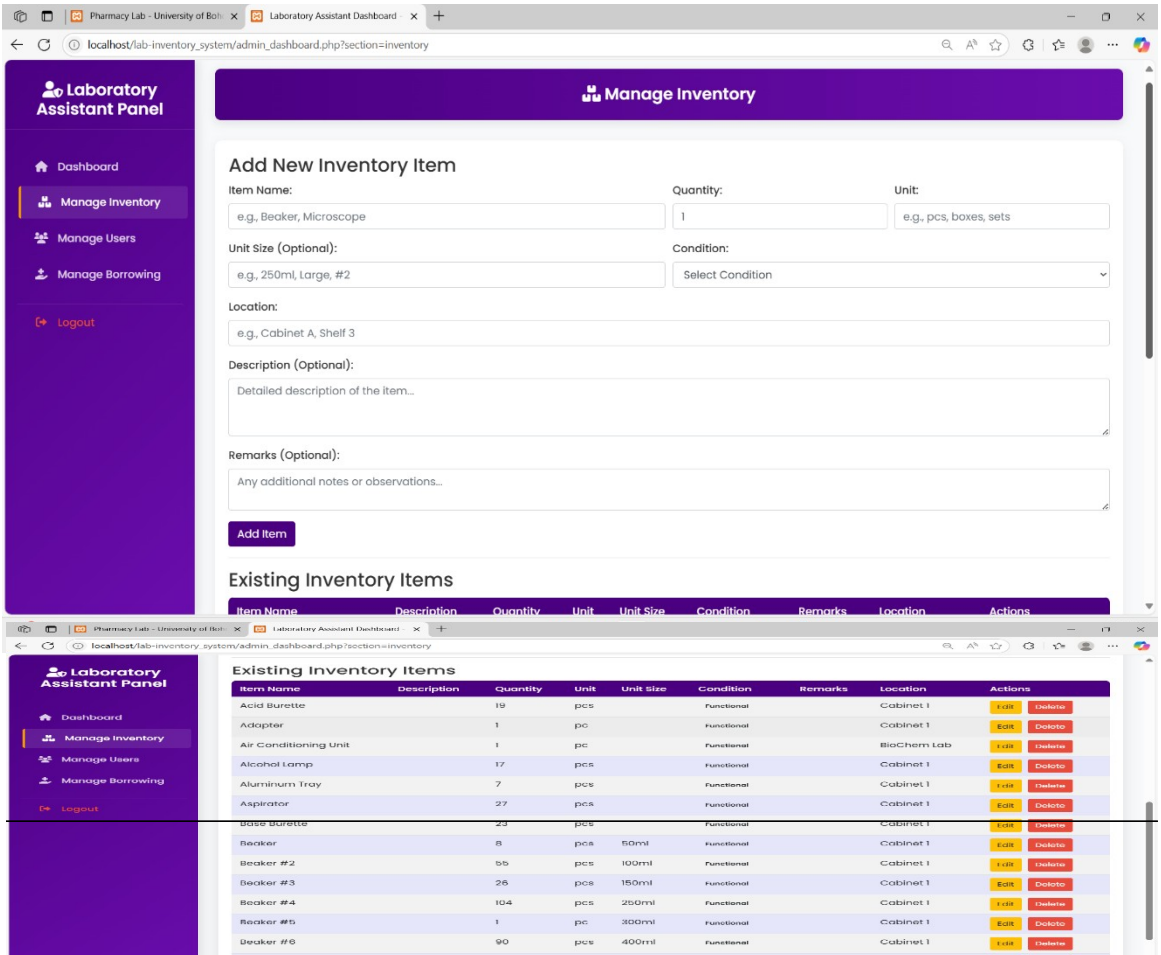
Admins can add, edit, or delete lab items.

Fields:

- **Item Name**
- **Quantity**
- **Unit (pcs, boxes)**
- **Unit Size (optional)**
- **Condition**
- **Storage Location**
- **Description and Remarks (optional)**

Displays:

A live table of all existing items with update/delete options.



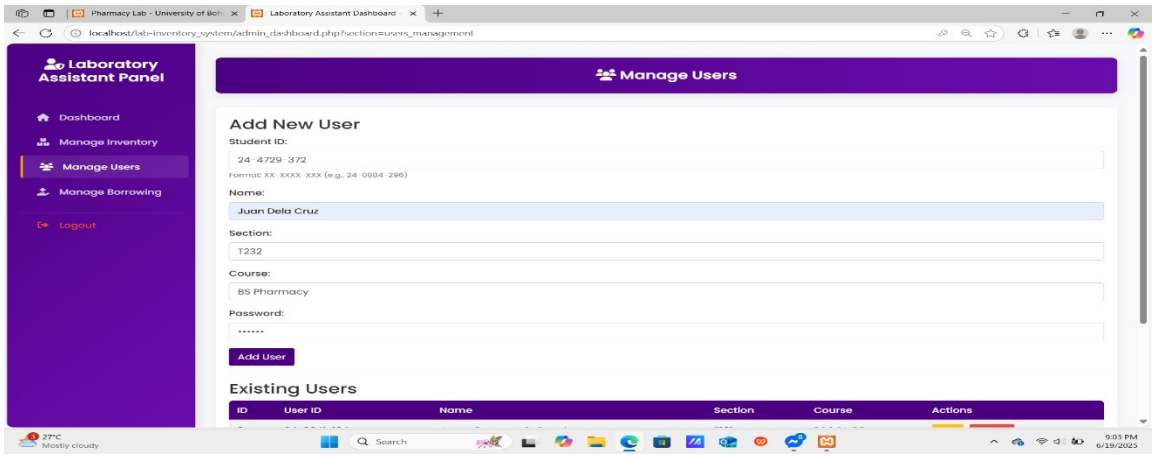
5. Manage Users

Description:

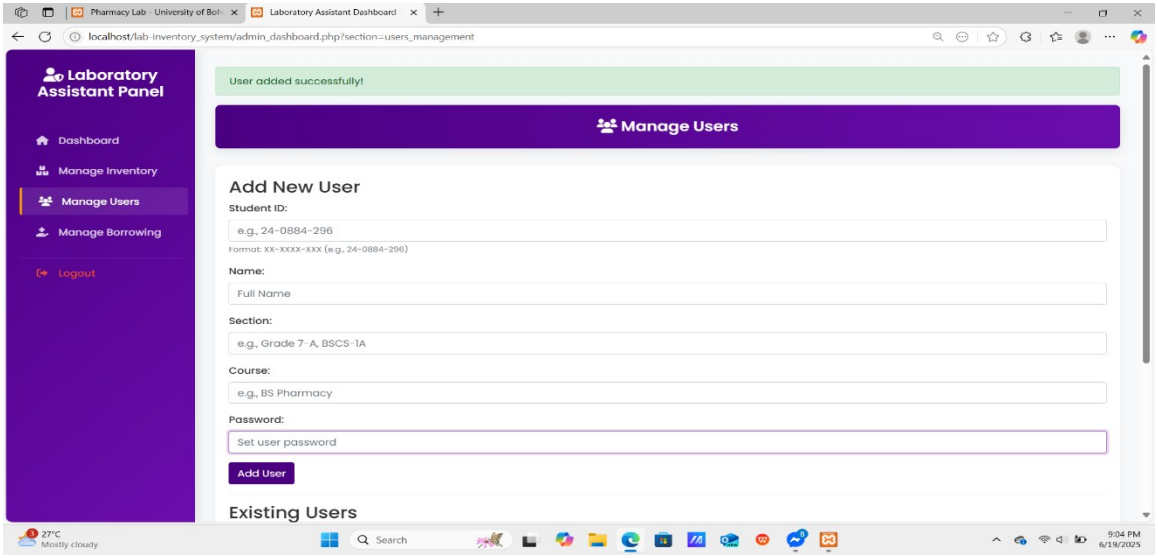
Displays all registered students. Admins can add or remove users if necessary.

Includes:

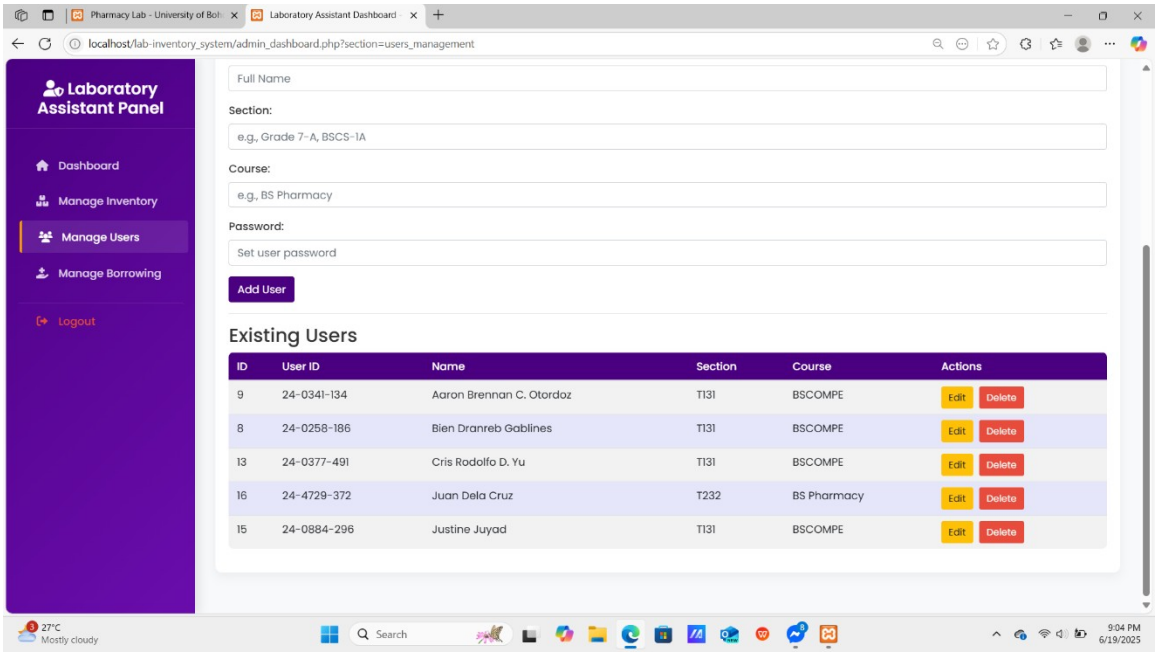
- **Student Name**
- **ID**
- **Section**
- **Course**
- **Actions (Edit/Remove/ADD)**



Add New User



Existing Users



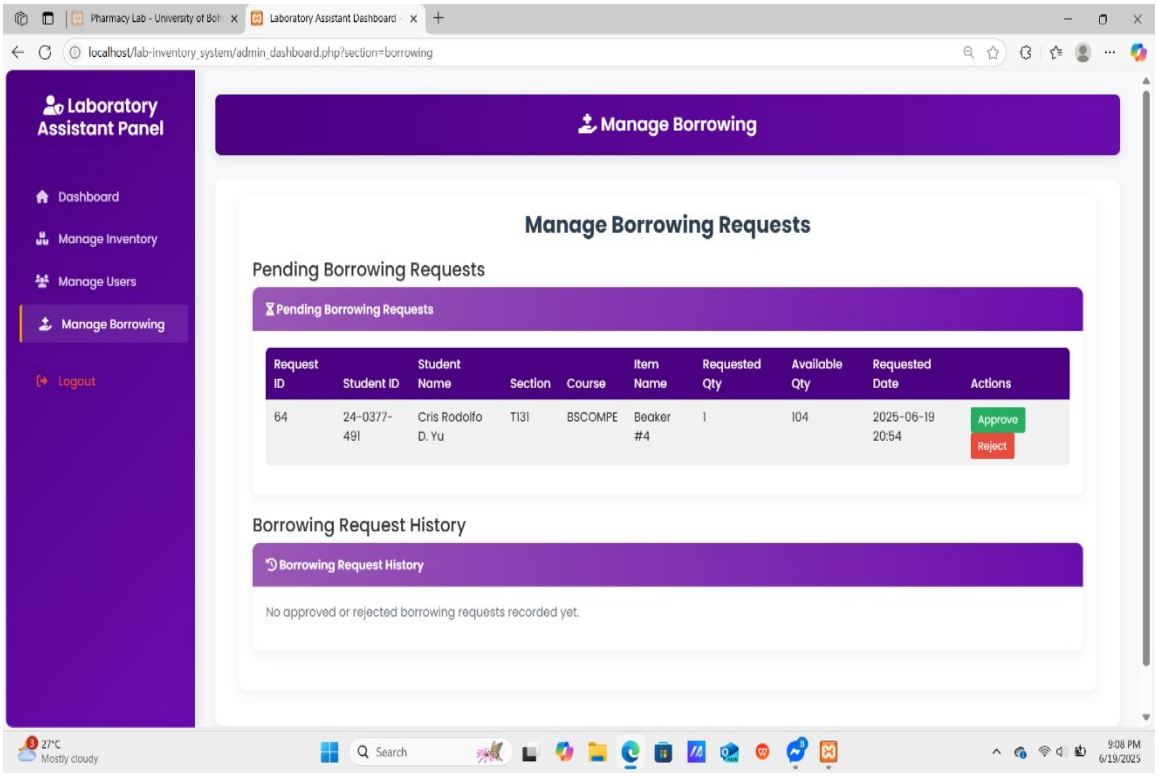
6. Manage Borrowing Requests

Description:

Displays all student borrowing requests for review.

Admin Options per Request:

- **Approve (if stock is sufficient)**
- **Decline (if invalid or unavailable)**
- **View Borrow Details**



7. Return Process (with Physical Checking)

Description:

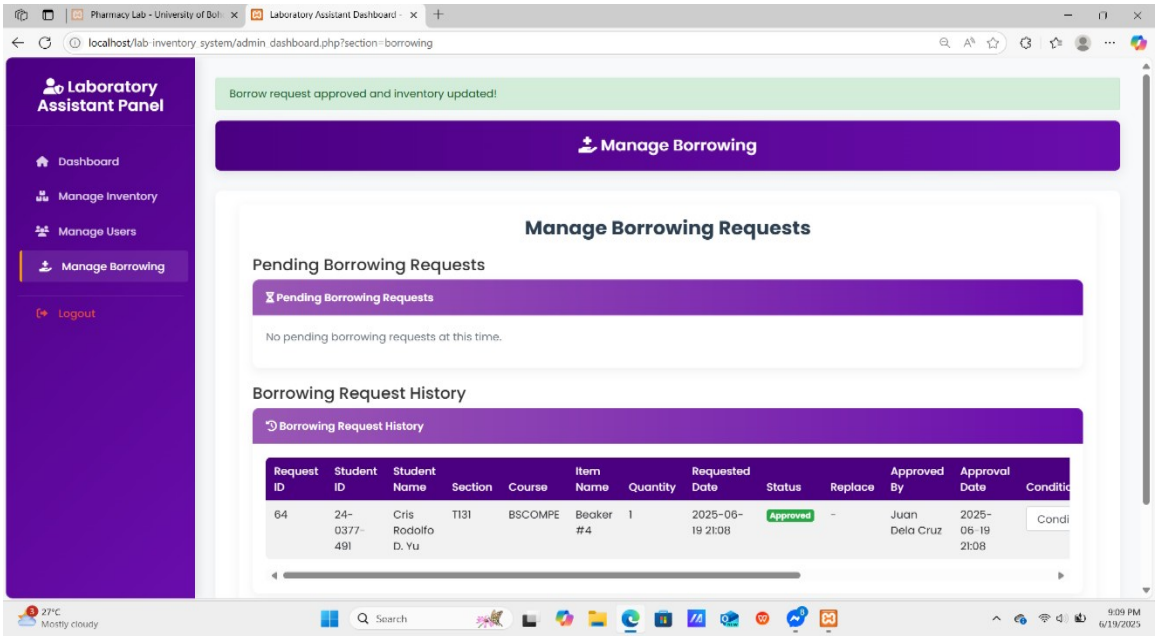
Admin confirms return of the borrowed item in person after checking its physical condition.

Steps:

- **Student presents the item in person**
- **Admin inspects it**
- **If item is in good condition, admin updates the request status to Returned**
- **Optionally logs remarks if item is damaged**

Purpose:

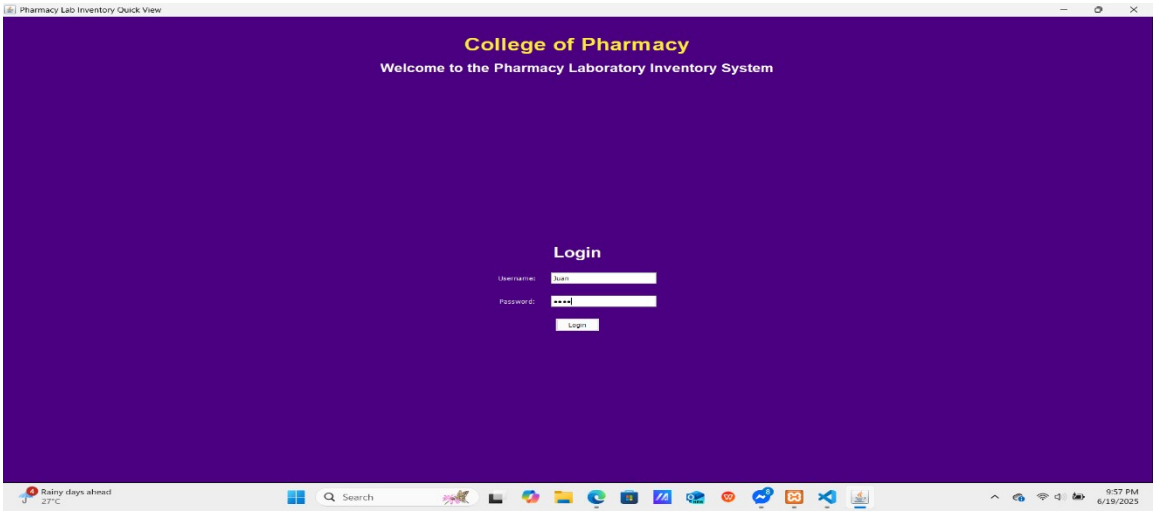
Prevents automatic returns and ensures physical validation.



JAVA INTERFACE

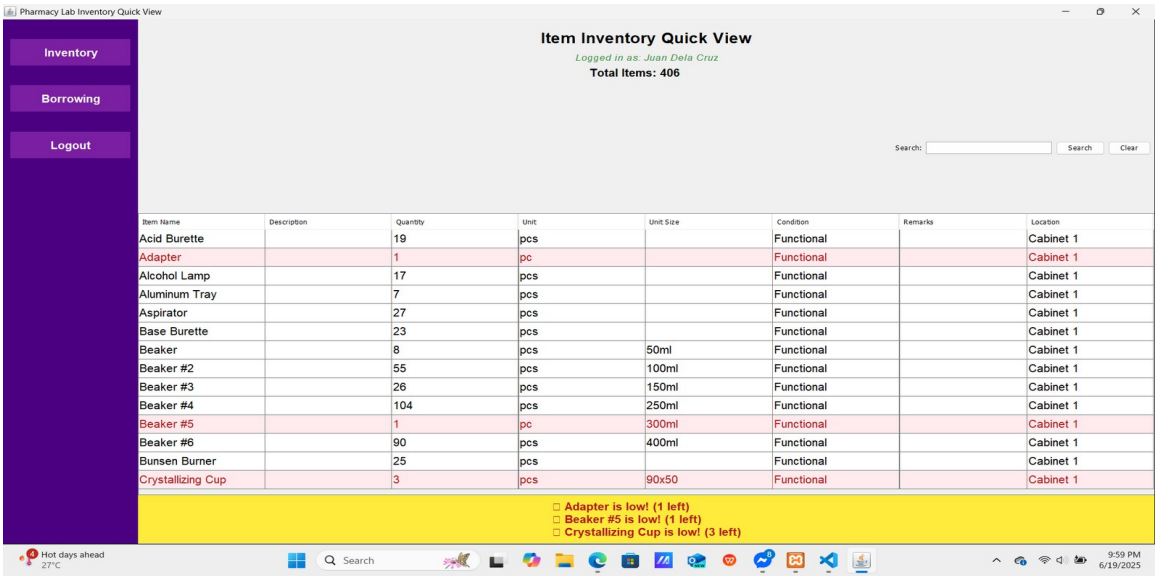
1. Admin Login

- **Verifies admin credentials from the admins table**
- **No registration; login only**



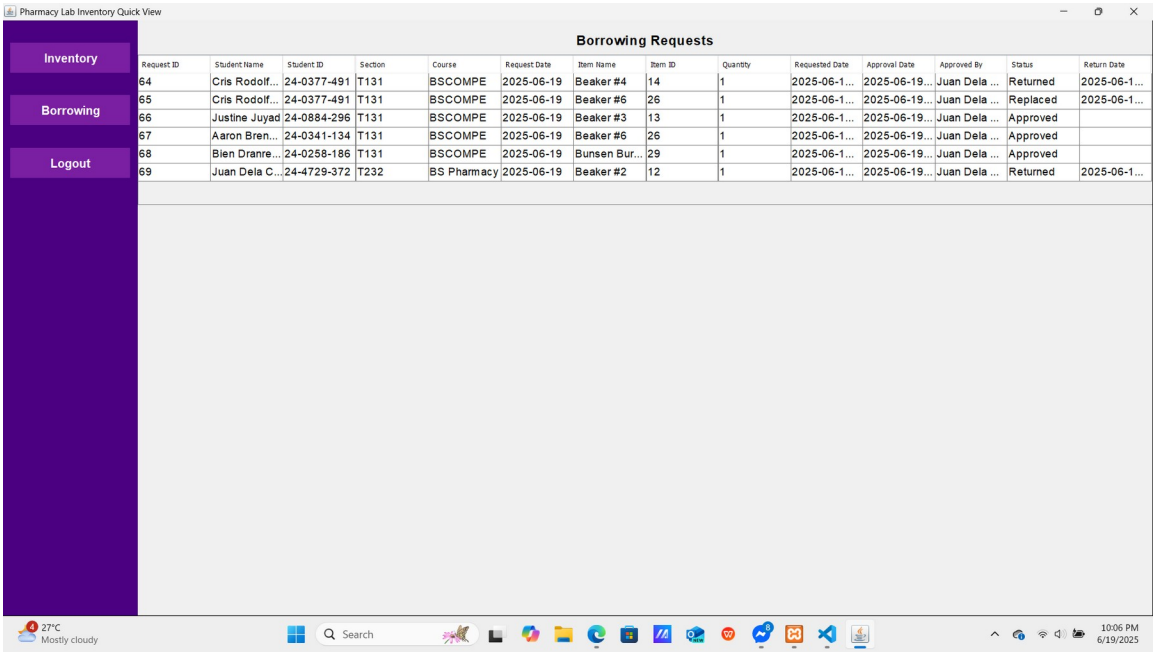
2. Inventory Overview

- **Fetches data from the inventory table**
- **Lists all items with name, quantity, condition**
- **Automatically flags items with quantity ≤ 5 as Low Stock**

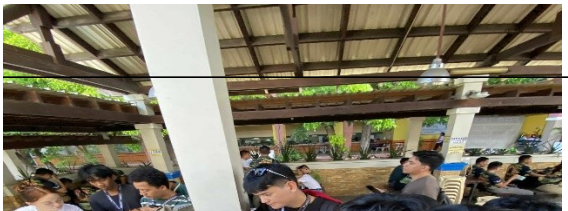


3. Borrowing History

- **Fetches data from borrow_requests**
- **Shows requests with status: Approved or Returned**
- **Displays: Item, Quantity, Student ID, Date, Status**

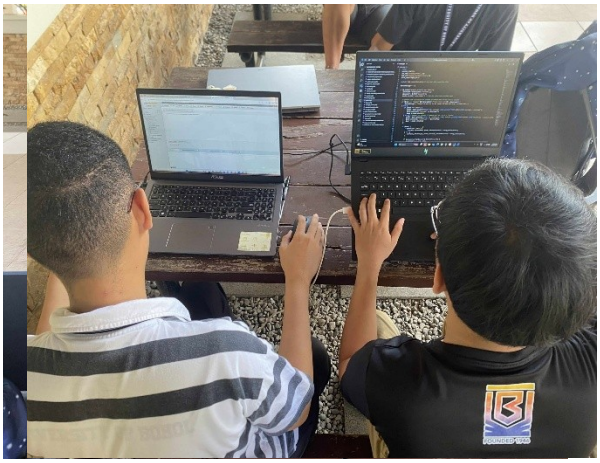


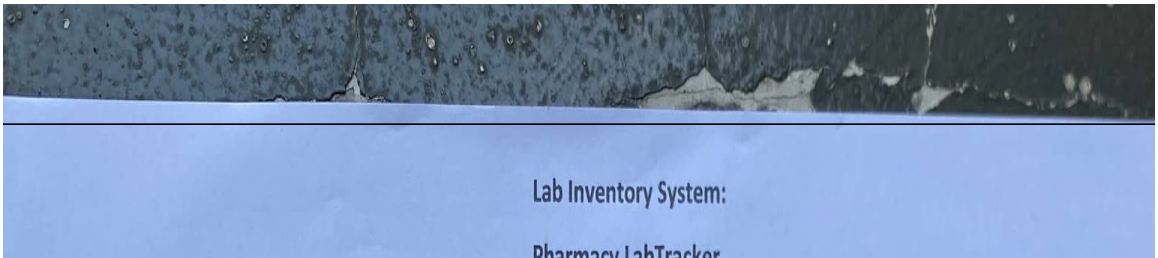
Checking Records Picture






PLEASE
CLEAN AS
YOU GO
SPOON & FORK
AREA

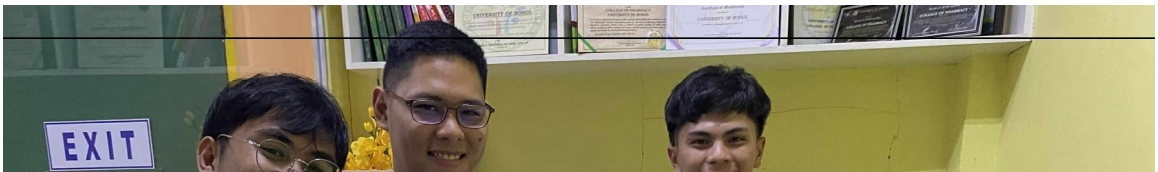


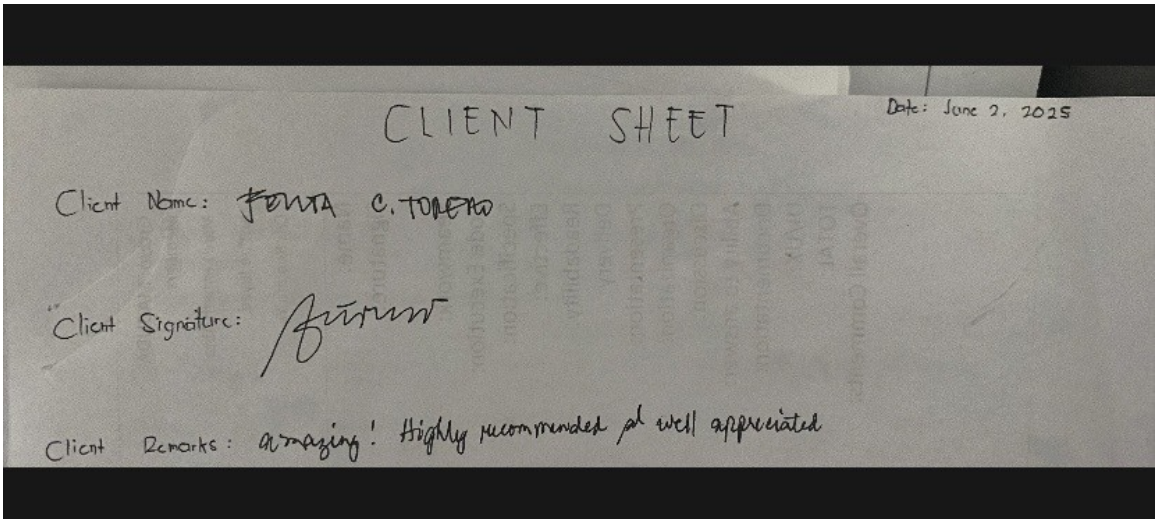

Lab Inventory System:

Group Journal during Project Development Picutre



Presenting the system of the Client Video





Youtube Video Url: https://www.youtube.com/watch?v=B37g5zp-RlM